

TD 2

Part 1 :

```
• @Henri127 →/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
```

1. Which function reads JSON?

load_instruments(), which uses json.load()

2. Which function reads CSV?

load_prices(), which uses csv.DictReader

3. Which function selects one ticker?

filter_prices(prices, ticker)

4. Which function coordinates execution?

main()

5. Where are the input files stored?

in data/sample/, DATA_DIR points to it. instruments.json holds the metadata and prices.csv holds the prices

Part 2 :

```

@Henri127 →/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ cat data/sample/instruments.json
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}

```

```

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Observations
AAPL: 21
SP500: 21
<class 'dict'>
dict_keys(['instrument', 'benchmark'])

```

```

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Observations
AAPL: 21
SP500: 21
<class 'dict'>
dict_keys(['instrument', 'benchmark'])
AAPL
Apple Inc.
USD
SP500
S&P 500

```

Part 3 :

```
@Henri127 →/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ head data/sample/prices.csv
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
```

```
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Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
<class 'list'>
<class 'dict'>
{'date': '2026-09-01', 'ticker': 'AAPL', 'open': '249.20', 'high': '251.50', 'low': '248.00', 'close': '250.00', 'volume': '38000000'}
```

```
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Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
<class 'list'>
<class 'dict'>
{'date': '2026-09-01', 'ticker': 'AAPL', 'open': '249.20', 'high': '251.50', 'low': '248.00', 'close': '250.00', 'volume': '38000000'}

250.00
<class 'str'>
```

Part 4 :

```
@Henri127 →/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ python src/main.py
=== MarketPulse ===
```

Instrument

AAPL - Apple Inc.

Last price: 266.20 USD

Benchmark

SP500 - S&P 500

Last level: 6742.00

Period: 1 month

Interval: Daily

Observations

AAPL: 21

SP500: 21

21

21

```
src > main.py
```

```
25 def filter_prices(prices, ticker):
26     #return [row for row in prices if row["ticker"] == ticker]
27     result = []
28
29     for row in prices:
30         if row["ticker"] == ticker:
31             result.append(row)
32     return result
```

Problems Output Debug Console Terminal Ports

```
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```

Instrument

AAPL - Apple Inc.

Last price: 266.20 USD

Benchmark

SP500 - S&P 500

Last level: 6742.00

Period: 1 month

Interval: Daily

Observations

AAPL: 21

SP500: 21

21

21

Part 5 :

```

27
28 def get_first_close(prices):
29     return float(prices[0]["close"])
30
31 def get_last_close(prices):
32     return float(prices[-1]["close"])
33
34 def main():

```

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Benchmark

SP500 - S&P 500

Last level: 6742.00

Period: 1 month

Interval: Daily

Observations

AAPL: 21

SP500: 21

AAPL: 250.0

SP500: 6600.0

AAPL: 266.2

SP500: 6742.0

```

print(f"{instrument['ticker']}: {get_first_close(instrument_prices)}")
print(f"{benchmark['ticker']}: {get_first_close(benchmark_prices)}")
print(f"{instrument['ticker']}: {get_last_close(instrument_prices)}")
print(f"{benchmark['ticker']}: {get_last_close(benchmark_prices)}")

```

Part 6 :

```

34 def display_market_summary(asset, prices, show_currency=True):
35     print(f"{asset['ticker']} - {asset['name']}")
36     print(f"Observations : {len(prices)}")
37     f_close=get_first_close(prices)
38     if (show_currency==False):
39         print(f"First Close : {get_first_close(prices)}")
40         print(f>Last Close : {get_last_close(prices)}")
41     else:
42         print(f"First Close : {get_first_close(prices)} USD")
43         print(f>Last Close : {get_last_close(prices)} USD")

```

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```

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=== MarketPulse ===

```

```

Market configuration
Period: 1 month
Interval: Daily

```

```

Instruments
AAPL - Apple Inc.
Observations : 21
First Close : 250.0 USD
Last Close : 266.2 USD

```

```

Benchmark
SP500 - S&P 500
Observations : 21
First Close : 6600.0
Last Close : 6742.0

```

```

print(("====="))
display_market_summary(instrument,instrument_prices)
display_market_summary(benchmark,benchmark_prices)

```

Part 8 :

```

@Henri127 →/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) $ python src/main.py
=== MarketPulse ===

```

```

Market configuration
Period: 1 month
Interval: Daily

```

```

Instruments
AAPL - AAPL
Observations : 21
First Close : 250.0 USD
Last Close : 266.2 USD

```

```

Benchmark
SP500 - SP500
Observations : 21
First Close : 6600.0 USD

```

Optional 1 :

```
40 def get_first_date(prices):
41     return prices[0]["date"]
42
43 def get_last_date(prices):
44     return prices[-1]["date"]
45
46 def main():
47     instruments = load_instruments()
48     prices = load_prices()
```

Problems Output Debug Console Terminal Ports

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SP500 - SP500
Observations : 21
First Close : 6600.0 USD
Last Close : 6742.0 USD

2026-09-01
2026-09-30

Optional 2 :

```
76 count=0
77 for i in range(0,len(instrument_prices)):
78     if (int(instrument_prices[i]["volume"]) > 0) :
79         count+=1
80 print(count)
```

Problems Output Debug Console Terminal Ports

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Benchmark
SP500 - SP500
Observations : 21
First Close : 6600.0 USD
Last Close : 6742.0 USD

2026-09-01
2026-09-30

21

Optional 3 :

```
52 def min_close(prices):
53     min=float(prices[0]["close"])
54     for i in range(1,len(prices)):
55         if(min>float(prices[i]["close"])):
56             min=float(prices[i]["close"])
57     return min
58
59 def max_close(prices):
60     max=float(prices[0]["close"])
61     for i in range(1,len(prices)):
62         if(max<float(prices[i]["close"])):
63             max=float(prices[i]["close"])
64     return max
65
66 def main():
67     instruments = load_instruments()
68     prices = load_prices()
```

Problems Output Debug Console **Terminal** Ports

@Henri127 →/workspaces/esilv-marketpulse-starter-g07-tSHA (Henri) \$ python src/main.py

Max instrument : 266.2
Min instrument : 249.8

Max benchmark : 6742.0
Min benchmark : 6598.0

Optional 4 :